

# MY457/MY557: Causal Inference for Observational and Experimental Studies

## Week 11: Difference-in-Differences 2

Daniel de Kadt

Department of Methodology  
LSE

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# Course Outline

- **Week 1:** The potential outcomes framework
- **Week 2:** Randomized experiments
- **Week 3:** Estimation under selection on observables I
- **Week 4:** Estimation under selection on observables II
- **Week 5:** Estimation under selection on observables III
- *Week 6: Reading week*
- **Week 7:** Instrumental variables I
- **Week 8:** Instrumental variables II
- **Week 9:** Regression discontinuity
- **Week 10:** Difference-in-differences I
- **Week 11:** Difference-in-differences II

# Difference-in-Differences So Far

Last week we studied the **canonical 2-period difference-in-differences** (DiD) design, with a brief foray into a special 3-period case.

Identification and estimation was reasonably straightforward:

- Key identification assumption is **parallel trends**, plus **no anticipation**
- Use either a plug-in or regression-based estimator

However, we often encounter DiD settings that are **more complex**:

- More than 2 time periods
- Treatment is assigned variably over time
- Treatment effects are heterogeneous (over e.g. time)
- Treatment is non-binary

Today we will consider **identification and estimation** in such settings.

- 1 Motivating Example
- 2 Fixed Effects
- 3 Variable Treatment Timing
- 4 Multi-Period Designs with Heterogeneous Treatment Effects
- 5 Synthetic Control Method Primer

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# Minimum Wages and Employment

Do higher minimum wages decrease employment?

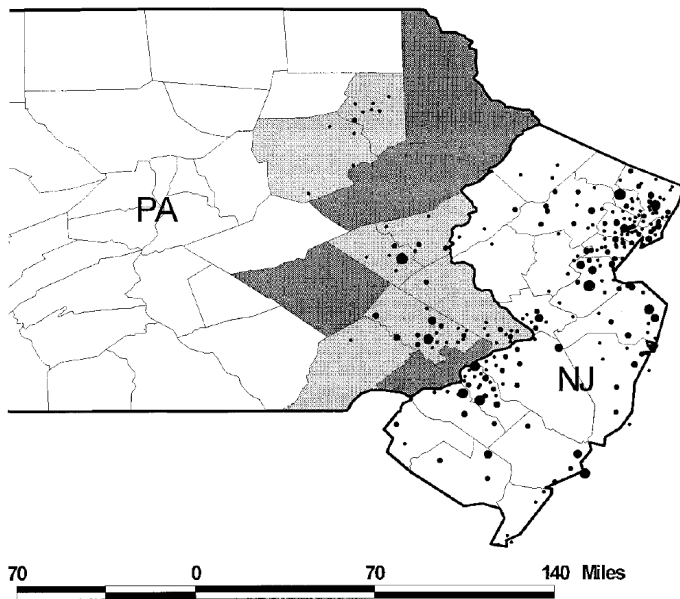
Card and Krueger (1994) consider impact of New Jersey's 1992 **minimum wage increase** from \$4.25 to \$5.05 per hour.

Compare employment in 410 fast-food restaurants in New Jersey and eastern Pennsylvania before and after the rise.

Survey data on wages and employment from **two waves**:

- Wave 1: March 1992, one month **before** the minimum wage increase
- Wave 2: December 1992, eight months **after** increase

# Minimum Wages and Employment



# Minimum Wages and Employment

| Variable                                                | Stores by state  |                 |                  |
|---------------------------------------------------------|------------------|-----------------|------------------|
|                                                         | PA<br>(i)        | NJ<br>(ii)      | Difference,      |
|                                                         |                  |                 | NJ – PA<br>(iii) |
| 1. FTE employment before,<br>all available observations | 23.33<br>(1.35)  | 20.44<br>(0.51) | - 2.89<br>(1.44) |
| 2. FTE employment after,<br>all available observations  | 21.17<br>(0.94)  | 21.03<br>(0.52) | - 0.14<br>(1.07) |
| 3. Change in mean FTE<br>employment                     | - 2.16<br>(1.25) | 0.59<br>(0.54)  | 2.76<br>(1.36)   |

Treatment: Increase of minimum wage in New Jersey

**Difference 1**: Pre-minimum wage - Post-minimum wage

**Difference 2**: New Jersey- Pennsylvania

# Minimum Wage and Employment

Card & Krueger (1994) found a **positive effect of minimum wage on employment**.

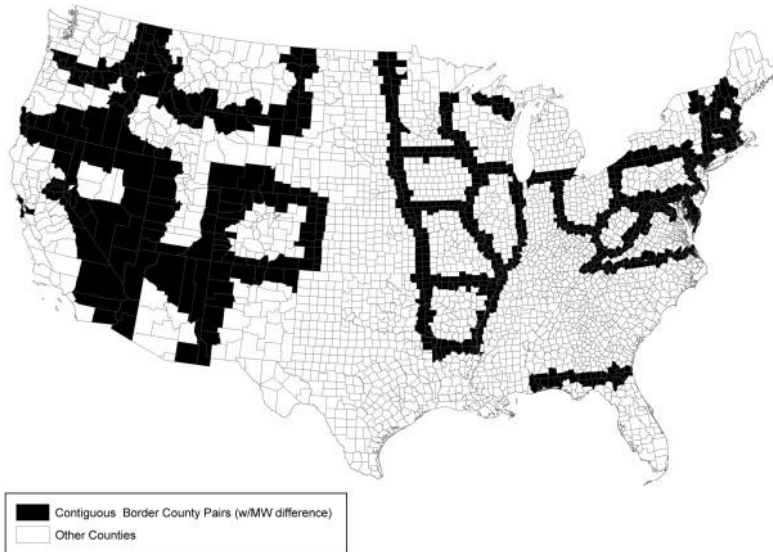
Card & Krueger (2000) revisited this design and setting with better data, and found **no effect** either way.

Lots of debate followed – many papers reconsidered this question using a more general approach: Leveraging cross- and within-state variation throughout the USA. They largely found **negative effects** on employment.

Dube, Lester, and Reich (2010) revisit this debate:

- Find all cross-state-border changes in MW policies (1990 - 2006)
- Collect earnings and employment data for every county in the USA in this time period.
- Generalize the DiD case study approach.

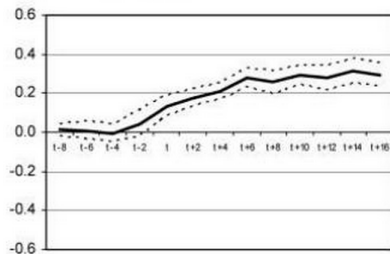
# Variation in Space



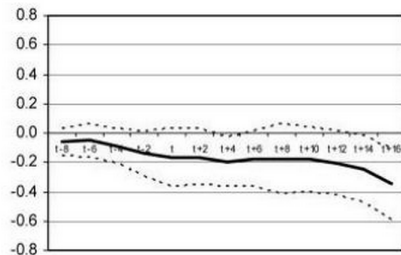
# Estimated Dynamic Effects – Entire Sample

## Ln Earnings

### 1. All County Sample, Common Period Effects

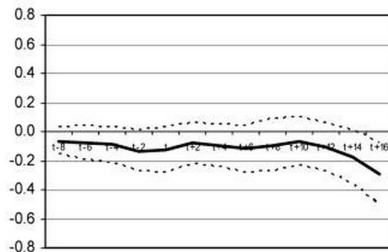
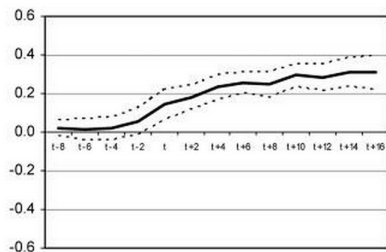


## Ln Employment



# Estimated Dynamic Effects – Border Sample Only

## 5. Contiguous Border County-Pair Sample, Common Period Effects



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Synthetic Control Method Primer

# Fixed Effects Estimation and Difference in Differences

Recall the **additive linear model for panel data** with 2 periods:

$$Y_{it}(\mathbf{z}) = \alpha_i + \gamma \mathbf{t} + \tau \mathbf{z} + \varepsilon_{it}$$

where

- $i \in \{0, \dots, N\}$ : Unit indicator
- $\mathbf{t} \in \{0, 1\}$ : Time indicator
- $Y_{it}(\mathbf{z})$ : potential outcome under treatment status  $\mathbf{z} \in \{0, 1\}$
- $\alpha_i$ : **time-invariant unit-specific unobserved effect**
- $\varepsilon_{it}$ : idiosyncratic error term
- $\tau$ : (homogeneous, constant) treatment effect of interest

In a 2-period design, we saw that the first-difference regression:

- Unbiasedly estimates  $\tau$  under parallel trends and no anticipation
- $\tau$  will coincide with  $\tau_{ATE}$  and  $\tau_{ATT}$  if assumed model is correct

**Fixed effects models** generalise this to  $\mathbf{t} > 2$ . We will first consider the **assumptions** required by different models (read: assumed data generating processes), then turn to **estimation**.

# Panel Data Notation and Setup

Let's introduce some generalised notation for panel data:

- **Unit**: Distinct analytical entities (e.g. people, countries, schools).
- **Observation**: One 'peek' at one unit (e.g. year, district, student).
- $y_{it}$ : Observed outcome for unit  $i$  in period  $t$
- $Z_{it} \in \{0, 1\}$ : Treatment indicator for unit  $i$  in period  $t$

Note 1: We are using **time** as our dimension of observation, but it could be other things – this goes beyond this lecture!

Note 2: Time-varying **covariates** can be included alongside  $Z_{it}$  – we omit them here for simplicity.

In a given panel, we observe a sample of  $i = 1, 2, \dots, N$  distinct units at  $t = 1, 2, \dots, T$  time periods (a 'balanced panel')

# Panel Data Notation and Setup

Collect variables for unit  $i$ :

$$\mathbf{y}_i = \begin{pmatrix} y_{i1} \\ \vdots \\ y_{it} \\ \vdots \\ y_{iT} \end{pmatrix}_{T \times 1} \quad \mathbf{z}_i = \begin{pmatrix} z_{i1} \\ \vdots \\ z_{it} \\ \vdots \\ z_{iT} \end{pmatrix}_{T \times 1}$$

And stack them for all units (a 'long panel'):

$$\mathbf{y} = \begin{pmatrix} \mathbf{y}_1 \\ \vdots \\ \mathbf{y}_i \\ \vdots \\ \mathbf{y}_N \end{pmatrix}_{NT \times 1} \quad \mathbf{z} = \begin{pmatrix} \mathbf{z}_1 \\ \vdots \\ \mathbf{z}_i \\ \vdots \\ \mathbf{z}_N \end{pmatrix}_{NT \times 1}$$

# Pooled OLS Model

Let's start by ignoring the panel structure entirely, and assume:

$$y_{it} = \tau Z_{it} + v_{it}, \quad t = 1, 2, \dots, T$$

with **composite error**  $v_{it} \equiv \alpha_i + \varepsilon_{it}$

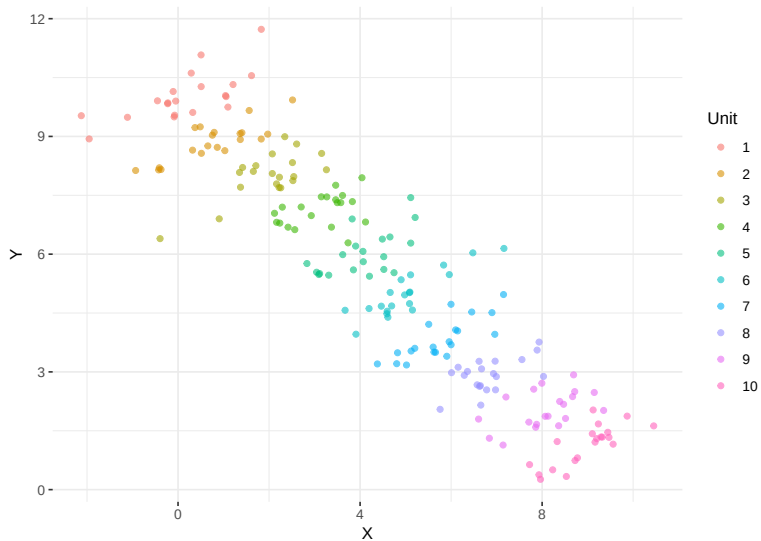
We could estimate this model directly using OLS. Our estimate of  $\tau$  will be unbiased and consistent under **contemporaneous exogeneity**:

$$\mathbb{E}[v_{it}|Z_{it}] = 0 \text{ for } t = 1, 2, \dots, T$$

Read: Since we control for nothing, we identify using **all variation** in  $Z_{it}$ . We must assume this variation is ignorable with respect to potential outcomes.

When might this be credible? Treatment is a **shock** that is **as-if random** with respect to **both units and time**.

# Pooled OLS: Illustration



# Unit Fixed Effects Model

Now consider a **unit fixed effects** assumed model:

$$y_{it} = \tau Z_{it} + \alpha_i + \varepsilon_{it}, \quad t = 1, 2, \dots, T$$

where  $\alpha_i$  captures **time-invariant unit differences**

We could estimate this model using OLS. Our estimate of  $\tau$  will be unbiased and consistent under **strict exogeneity**:

$$\mathbb{E}[\varepsilon_{it} | Z_{i1}, Z_{i2}, \dots, Z_{iT}, \alpha_i] = 0, \quad t = 1, 2, \dots, T$$

Read: Once we control for  $\alpha_i$ , we identify using **within-unit over-time variation** in  $Z_{it}$ . We assume this variation is (conditionally) ignorable.

When might this be credible? When treatment is a **shock** that is **as-if random** with respect to **time**.

If the assumed model is correct, this is a generalization of the **pre-post design**!

# (Unit) Fixed Effects Estimation

Three equivalent approaches to estimating  $\tau$  in the unit FE model:

1. The **LSDV estimator**:

- a. Regress  $y_{it}$  on  $Z_{it}$  and unit dummies

Note 1: Here we directly estimate  $\alpha_j$

Note 2: With large  $N$ , this can be computationally expensive

2. **Within** estimation:

- a. Create demeaned variables:  $\ddot{Z}_{it} \equiv Z_{it} - \bar{Z}_i$  and  $\ddot{y}_{it} \equiv y_{it} - \bar{y}_i$

- b. Regress  $\ddot{y}_{it}$  on  $\ddot{Z}_{it}$

Note: By within-demeaning we 'purge' the fixed effects  $\alpha_j$

3. **First differences** estimation (like last week):

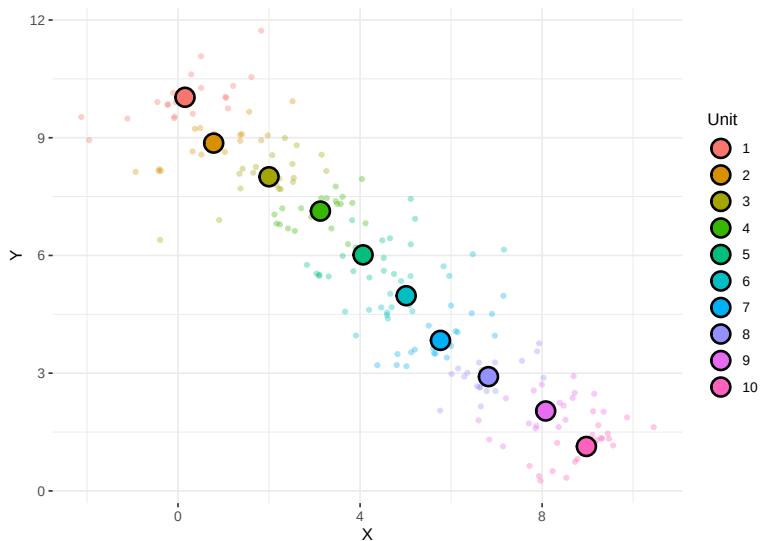
- a. Create differenced variables:  $\Delta Z_{it} = Z_{it} - Z_{i,t-1}$  and  $\Delta y_{it} = y_{it} - y_{i,t-1}$

- b. Regress  $\Delta y_{it}$  on  $\Delta Z_{it}$

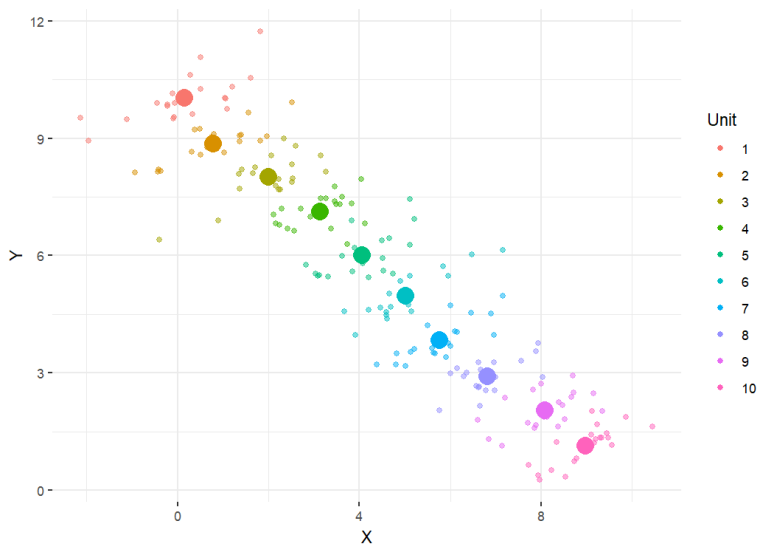
Note: First differencing also purges  $\alpha_j$

All three yield the same  $\widehat{\tau_{FE}}$ .

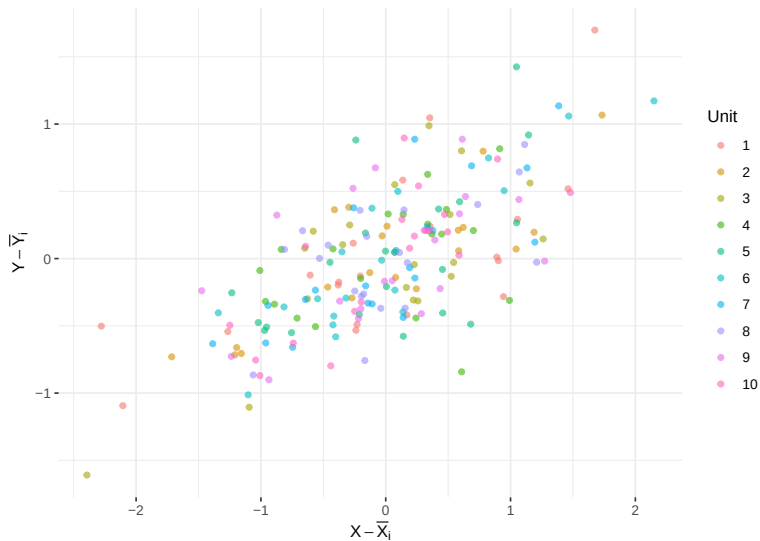
# Within Estimation: Unit Means



# Within Estimation: De-meaning



# Within Estimation: De-meaned OLS



# Adding Time Effects

Consider again our assumed 'unit fixed effects' model:

$$y_{it} = \tau Z_{it} + \alpha_i + \varepsilon_{it}, \quad t = 1, 2, \dots, T$$

Recall we assume on ignorable within-unit temporal variation in  $Z_{it}$ .

Typical violation: **common changes over time** that affect all units'  $y_{it}$  in the same way and are correlated with  $Z_{it}$  (e.g. **history** or **maturational** effects).

More realistic models might include **time effects**:

- linear time trends
- non-linear time trends
- unit-specific time trends
- **time fixed effects**

# Two-Way Fixed Effects Regression

Let's add **time fixed effects** to our assumed model:

$$y_{it} = \tau Z_{it} + \alpha_i + \delta_t + \varepsilon_{it}, \quad t = 1, 2, \dots, T$$

where

- $\alpha_i$  captures time-invariant unit differences
- $\delta_t$  captures common time-specific differences

This is the **two-way fixed effects** (TWFE) model.

Once we account for  $\alpha_i$  and  $\delta_t$ , we identify using **idiosyncratic variation** in  $Z_{it}$ . We assume this variation is ignorable.

When might this be credible? Treatment is a **shock** whose **timing** varies across units for **idiosyncratic** reasons. If the assumed model is correct, our estimate of  $\tau$  is **generalized difference-in-differences**.

Estimation as before, extending each approach by including time dummies. In R, use `plm` or `fixest` (preferred).

# Dynamic Two-Way Fixed Effects

We can specify a TWFE model allowing **dynamic (time-varying)** treatment effects:

$$y_{it} = \alpha_i + \delta_t + \sum_{r \neq 0} \mathbf{1}[R_{it} = r] \tau_r + \varepsilon_{it}$$

where

- $\alpha_i$  captures time-invariant unit differences
- $\delta_t$  captures common changes over time
- $R_{it}$  is the period relative to treatment for unit  $i$ , with  $R_{it} = 0$  denoting the last pre-treatment period (reference category)
- $\tau_r$  is a relative-period treatment effect

Read: This **event study** estimator allows for heterogeneous treatment effects of a specific form: **constant  $\tau_r$  across treatment cohorts**.

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# Variable Treatment Timing

Multi-period treatment regimes usually vary over two dimensions:

- **Uniform vs. Staggered**: Does treatment occur simultaneously, or over time?
- **Absorbing vs. Non-absorbing**: Once treatment occurs, can it switch off?

With anything other than uniform and absorbing treatment timing, TWFE for DiD may not behave well. For synthesis, see:

- Baker, Larcker, and Wang (2022)
- Roth, Sant'Anna, Bilinski, and Poe (2023)

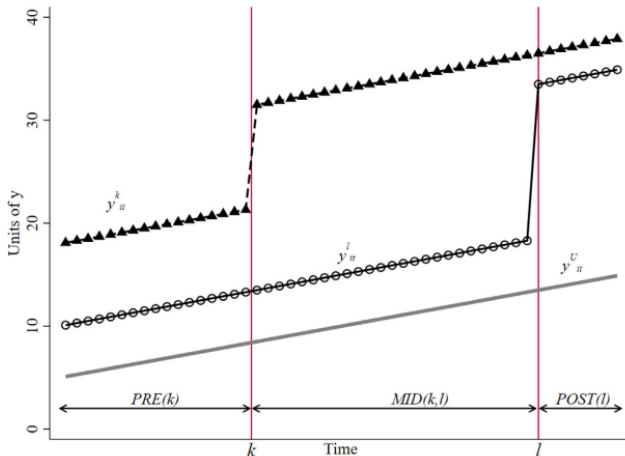
Short of further assumptions, the estimand targeted by TWFE is **not easily interpretable**  $\rightsquigarrow$  it is a weighted average of many different treatment effects.

These weights can be negative (!), are generally non-intuitive, and can potentially severely mislead (e.g. sign-flips).

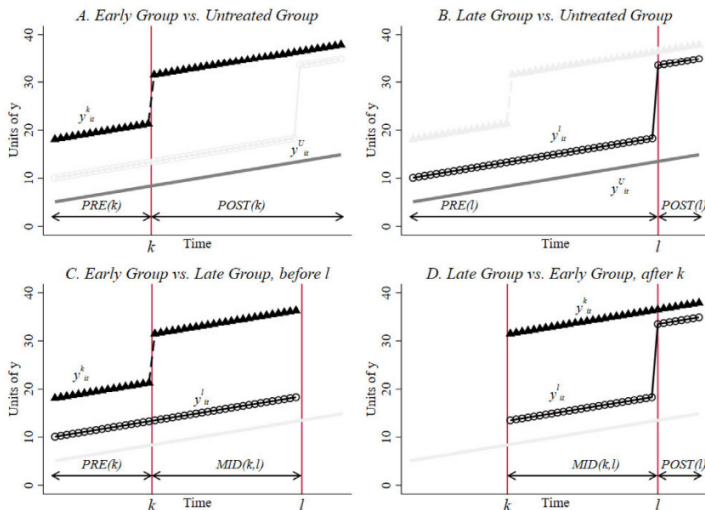
## $\widehat{\tau}_{TWFE}$ Decomposition

To see this, we can **decompose**  $\widehat{\tau}_{TWFE}$ . We focus on Goodman-Bacon (2021).

Define three groups: never treated (U), early treated (k), and late treated (l)



# $\widehat{\tau}_{TWFE}$ Decomposition



$\widehat{\tau}_{TWFE}$  is the weighted average of these four 2x2 treatment effects.

## $\widehat{\tau}_{TWFE}$ Decomposition

$$\widehat{\tau}_{TWFE} = \sum_{k \neq U} s_{kU} \hat{\tau}_{kU}^{2 \times 2} + \sum_{k \neq U} \sum_{\ell > k} [s_{kl}^k \hat{\tau}_{k\ell}^{2 \times 2, k} + s_{kl}^l \hat{\tau}_{k\ell}^{2 \times 2, \ell}]$$

where

- $\hat{\tau}^{2 \times 2}$  are different 2x2 estimators
- $s$  are estimator-specific weights

The weights are a function of three things:

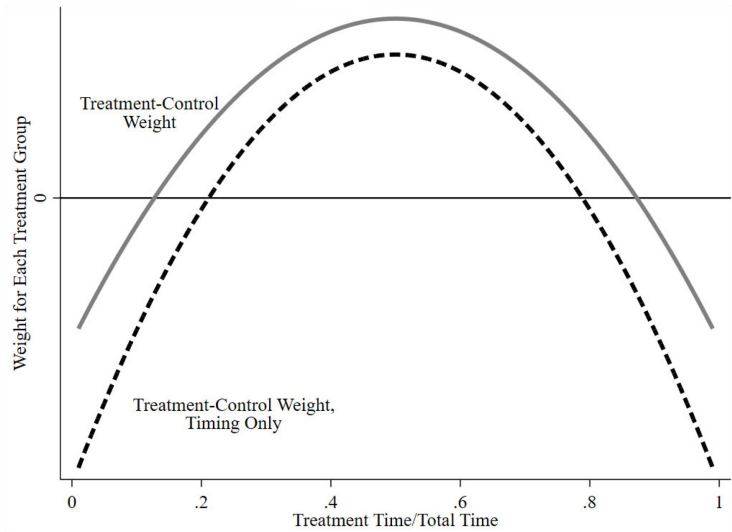
1. 2x2 subsample size (higher,  $\uparrow$  weight)
2. Ratio of treated to control (closer to 0.5,  $\uparrow$  weight)
3. Relative timing of treatment (more central,  $\uparrow$  weight)

Problem: Some of these comparisons may be **forbidden** such that already-treated units used as controls after they are treated.

Intuition: We subtract off changes in untreated outcomes **and treatment effects**.

## $\tau_{TWE}$ Decomposition

Weighting is heavily dependent on timing:



# Key Assumptions for TWFE as Generalised DiD

This decomposition reveals the assumptions under which **traditional TWFE** might be trusted with **multi-period panel data** and a **DiD** design.

A **causal DiD** interpretation with **TWFE** requires either:

1. Parallel trends, no anticipation, and homogenous  $\tau$

Or:

2. Parallel trends, no anticipation, and uniform timing (constant  $\tau$  over time)

Quickly explore how much of a problem this may be using **bacondecomp** in R.

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# Two Classes of 'Modern' Estimators

Multi-period DiD with **non-uniform (staggered or non-staggered) treatment timing** should be approached with **caution**.

Two general types of modern estimators that can help:

## 1. **Flexible matching and re-weighting** estimators:

- Make the 'right' comparisons only, weight appropriately, and recover  $\tau_{ATT}$ .
- Many estimators exist: Strehznev (2018), de Chaisemartin and D'Haultfœuille (2020), Sun and Abraham (2021), Imai and Kim (2021), Dube et al. (2023), de Chaisemartin and D'Haultfœuille (2024)
- We will focus on Callaway and Sant'Anna (2021)

## 2. **Counterfactual** estimators:

- Estimate only  $Y_0$ , thus avoiding forbidden comparisons, and recover  $\tau_{ATT}$ .
- Many estimators exist: Gobillon and Magnac (2016), Xu (2017), Borusyak et al. (2021), Gardner (2021), Wooldridge (2021)
- We will focus on Liu, Wang, and Xu (2022)

## Callaway and Sant'Anna (2021): Setup

Callaway and Sant'Anna (2021) study a DiD setting with **multiple time periods**, **staggered** treatment timing, there may be **heterogeneous**  $\tau$ , and parallel trends may hold only conditional on  $\mathbf{X}$ .

They begin by defining a new estimand, the **group-time ATT**:

$$\tau_{g,t}^{ATT} = \mathbb{E}[Y_t(1) - Y_t(0) | G_g = 1]$$

where

- there are  $T = t \in \{1, \dots, T\}$  time periods
- $G_g \in \{0, 1\}$  indicates whether a unit is first treated in period  $g$
- $Y_t(1)$  and  $Y_t(0)$  are potential outcomes under treatment and control, at  $T = t$

Intuitively, this has already brought us a long way. We can now reason in abstraction about **every** 2x2 comparison in our data.

## Callaway and Sant'Anna (2021): Identification

If **parallel trends** holds, this group-time ATT can be identified as:

$$\tau_{g,t}^{ATT} = \mathbb{E}[\underbrace{Y_t - Y_{g-1}}_{\text{Long difference}} | G_g = 1] - \mathbb{E}[\underbrace{Y_t - Y_{g-1}}_{\text{Long difference}} | C = 1]$$

where  $C \in \{0, 1\}$ , taking 1 if never treated (no forbidden comparisons!)

Read: For any treated cohort, at any time period, conduct a diff-in-diff where:

- Difference 1:  $T = t$  vs.  $T = g - 1$  (long difference)
- Difference 2:  $G_g$  (cohort) vs.  $C$  (never treated)

**Long difference** in  $Y$ : change in  $Y$  between any time period  $t$  and the **last pre-treatment period**,  $g - 1$ .

# Callaway and Sant'Anna (2021): Identification

If **conditional parallel trends** holds:

$$\tau_{g,t}^{ATT} = \mathbb{E} \left[ \underbrace{\left( \frac{G_g}{\mathbb{E}[G_g]} - \frac{\frac{p_g(X)C}{1-p_g(X)}}{\mathbb{E} \left[ \frac{p_g(X)C}{1-p_g(X)} \right]} \right)}_{\text{Weights}} \underbrace{(Y_t - Y_{g-1})}_{\text{Long difference}} \right]$$

$p_g(X) = P(G_g = 1|X, G_g + C = 1)$  is a propensity score

Read: Up-weight control units where  $p_g(X)$  is similar to the **group-specific treated units**

# Callaway and Sant'Anna (2021): Estimation

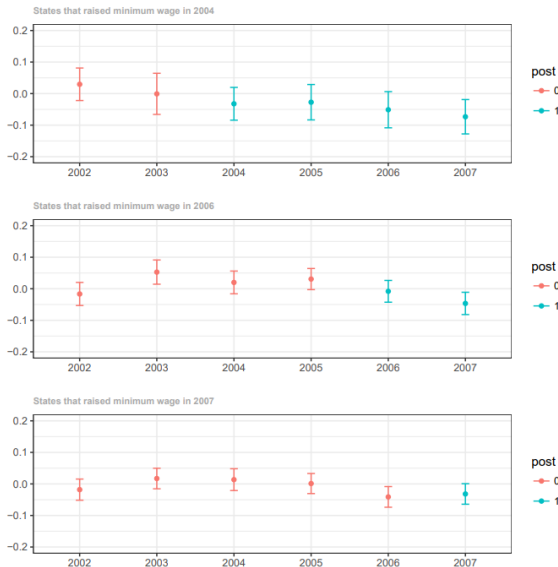
Estimation proceeds as follows:

1. Estimate  $\hat{p}_g$  for each group  $g$
2. Estimate  $\hat{\tau}_{g,t}^{ATT}$  by plugging in fitted values and observed  $Y$  into the (estimator-version) of the expression on the previous slide
3. Combine the estimated values of  $\hat{\tau}_{g,t}^{ATT}$  to retrieve quantities of interest

Some quantities of interest:

- Simple average of  $\hat{\tau}_{g,t}^{ATT}$  across  $t$  and  $g$
- Weighted average of  $\hat{\tau}_{g,t}^{ATT}$  weighting by group sizes
- Any other principled summary measure!

# Callaway and Sant'Anna (2021) on the Minimum Wage



## Liu, Wang, and Xu (2022): Setup

Liu, Wang, and Xu (2022) consider DiD settings with **multiple** time periods, **staggered** treatment timing that **may or may not be absorbing**, and there may be **heterogeneous**  $\tau$ .

Define the **estimand** of interest as:

$$\tau_{ATT} = \mathbb{E}[Y_{it}(1) - Y_{it}(0) \mid Z_{it} = 1, C_i = 1]$$

where

- $Z_{it}$  is our DiD treatment indicator
- $C_i$  is an indicator for 'ever treated' units
- $Y_{it}(1)$  and  $Y_{it}(0)$  are potential outcomes under treatment and control

Idea: Estimate **only**  $Y_{it}(0)$  using pre-treatment data, taking  $Y_{it}(1)$  as missing.

Estimate  $\tau_{ATT}$  by taking differences between  $Y(1)$  and  $Y(\hat{0})$ .

Note: This is a philosophical departure from TWFE!

# Liu, Wang, and Xu (2022): Estimation

Authors offer three estimators:

- FEct Estimator:

$$Y_{it}(0) = \mathbf{x}_{it}^{\top} \boldsymbol{\beta} + \alpha_i + \delta_t + \varepsilon_{it}$$

- IFect Estimator:

$$Y_{it}(0) = \mathbf{x}_{it}^{\top} \boldsymbol{\beta} + \alpha_i + \delta_t + \lambda_i' \mathbf{f}_t + \varepsilon_{it}$$

- MC Estimator:

$$\mathbf{Y}(0) = \mathbf{x}_{it}^{\top} \boldsymbol{\beta} + \mathbf{L} + \boldsymbol{\varepsilon}$$

## Liu, Wang, and Xu (2022): Interactive Fixed Effects

Let's look briefly at the **interactive fixed effects** estimator.

Within the TWFE setup, replace  $\alpha_i + \delta_t$  with  $\alpha_i + \delta_t + \lambda_i^\top \mathbf{f}_t$  (Bai, 2009):

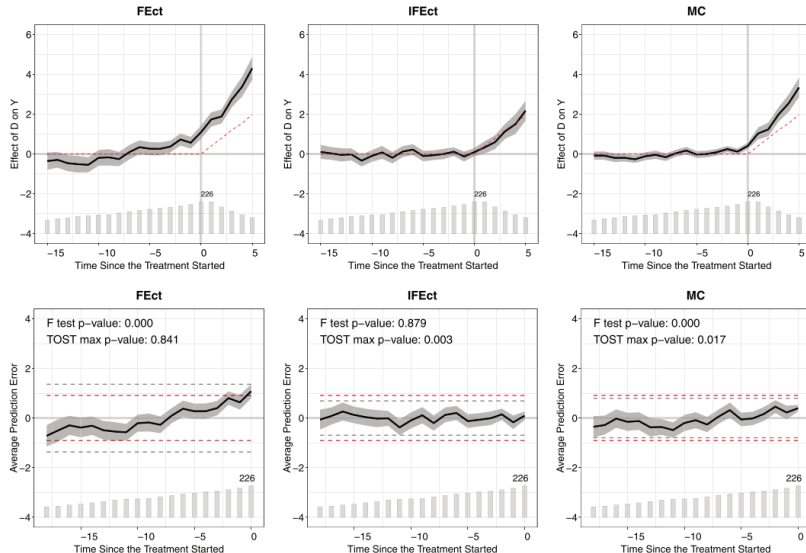
- $\mathbf{f}_t$ : unobserved **common factors** – time-varying, shared by all units
- $\lambda_i$ : unit-specific **factor loadings** – how sensitive unit  $i$  is to each factor

Read: If parallel trends fails because time effects are not actually common, IFECt finds those differences in the pre-treatment data and adjusts the counterfactual.

Note: This does not require covariates (though they can be used). Instead amounts to finding a low-dimensional representation of  $\mathbf{Y}(\mathbf{0})$  in the pre-period.

Estimation uses an **EM algorithm** on untreated observations to jointly recover  $\hat{\mathbf{f}}_t$  and  $\hat{\lambda}_i$ ; the number of factors  $r$  is selected via **cross-validation**.

# Liu, Wang, and Xu (2022): Simulated Example



# Some Practical Advice

For computation, most of these routines are well packaged in most software:

1. For Callaway & Sant'Anna package `{did}` in R.
2. For Liu et al, package `{fect}` in R.
3. See Asjad Naqvi's summary: <https://asjadnaqvi.github.io/DiD/>

Be aware, however:

- TWFE is often okay! Try a number of estimators, report honestly.
- 'Modern' estimators have low power (see Chiu et al, 2026, and Weiss, 2026)
- Low power may mean quite variable point estimates.
- Focus on **research design and falsification** first!

**Design precedes estimation!**

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# Synthetic Control Method: Primer

DiD requires parallel trends in the expected value of potential outcomes.  
Generally cannot help where there are **time- and unit-varying confounders**.

Synthetic Control Methods (SCM) take a **different approach**:

1. Find  $\mathbf{W}^*$ , an  $N - 1$  length vector of unit weights that minimizes  $||\mathbf{X}_1 - \mathbf{X}_0 \mathbf{w}||$  for  $\mathbf{X}_1$  a matrix of pre-treatment **outcomes and covariates** for the treated unit, and  $\mathbf{X}_0$  likewise for the control.

Read: Find the weighted combination of control units that best matches – in **both levels and trend** – the treated unit in the pre-treatment period.

This weighted set of control units is the **synthetic control**.

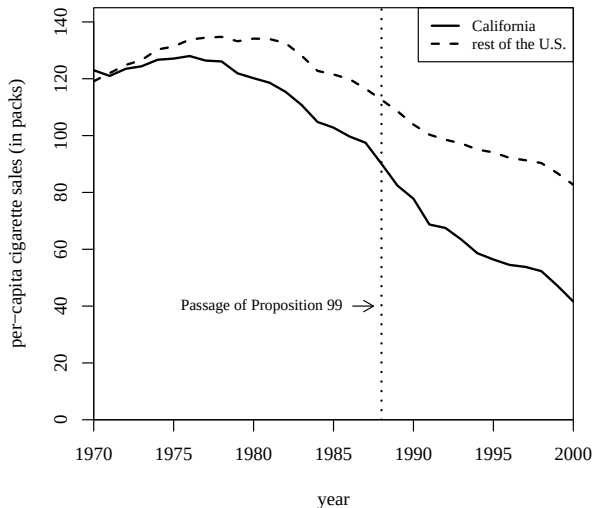
2. An approximately unbiased estimator of the unit-specific effect in the post-treatment period is:

$$\hat{\tau}_{1t} = Y_{1t} - \sum_{j=2}^{J+1} w_j^* Y_{jt} \quad \text{for } t \in \{T_0 + 1, \dots, T\}$$

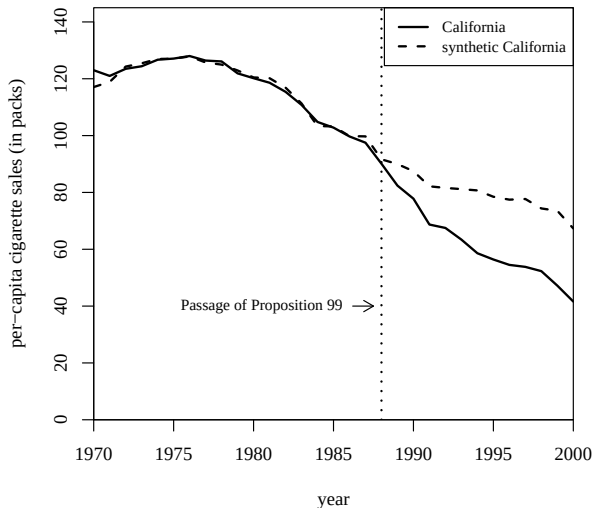
3. Inference uses placebos (roughly, randomization inference).

For primer on use, see Abadie (2020).

## SCM Example: California Prop 99 (Abadie et al. 2010)



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